

DATABASE CAR PARKING SYSTEM

EXPERIMENT NO. 8

TITLE

Subquery and Correlated Query

AIM

To perform subquery and correlated query operations on the Car Parking System database.

DESCRIPTION

A subquery is a query nested inside another query. The inner query provides values used by the outer query. A correlated subquery uses data from the outer query and is evaluated for each row of the outer query.

SUBQUERY

1. Which parking records have an amount greater than the average parking amount?

```
SELECT vehicle_id, slot_id, amount
FROM PARKING
WHERE amount > (SELECT AVG(amount) FROM PARKING);
```

2. Which vehicles have more than one parking record?

```
SELECT *
FROM VEHICLE
WHERE vehicle_id IN (
    SELECT vehicle_id
    FROM PARKING
    GROUP BY vehicle_id
    HAVING COUNT(*) > 1
);
```

CORRELATED SUBQUERY

3. Which staff member has joined recently and when?

```
SELECT staff_name, join_date
FROM STAFF s
WHERE join_date = (
    SELECT MAX(s2.join_date)
    FROM STAFF s2
);
```

4. List parking records whose amount is greater than the average amount for the same vehicle.

```
SELECT p.vehicle_id, p.slot_id, p.amount
FROM PARKING p
WHERE p.amount > (
    SELECT AVG(p2.amount)
    FROM PARKING p2
    WHERE p2.vehicle_id = p.vehicle_id
);
```

OUTPUTS

1) Parking records above the average parking amount.

```
+-----+-----+-----+
| vehicle_id | slot_id | amount |
+-----+-----+-----+
|          3 |      102 | 200.00 |
|          4 |      105 | 300.00 |
+-----+-----+-----+
2 rows in set (0.042 sec)
```

2) Vehicles having more than one parking record.

```
+-----+-----+-----+-----+
| vehicle_id | owner_name | vehicle_no | vehicle_type |
+-----+-----+-----+-----+
|          1 | Arun      | TN01AB1234 | Car          |
+-----+-----+-----+-----+
1 row in set (0.222 sec)
```

3) Staff member who joined recently and the joining date.

```
+-----+-----+-----+-----+
| vehicle_id | owner_name | vehicle_no | vehicle_type |
+-----+-----+-----+-----+
|          1 | Arun      | TN01AB1234 | Car          |
+-----+-----+-----+-----+
1 row in set (0.222 sec)
```

4) Parking records above the vehicle-wise average amount.

```
+-----+-----+-----+
| vehicle_id | slot_id | amount |
+-----+-----+-----+
|          1 |      101 | 120.00 |
+-----+-----+-----+
1 row in set (0.009 sec)
```

RESULT

The records from the Car Parking System database were displayed using Subquery and Correlated Subquery.